KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

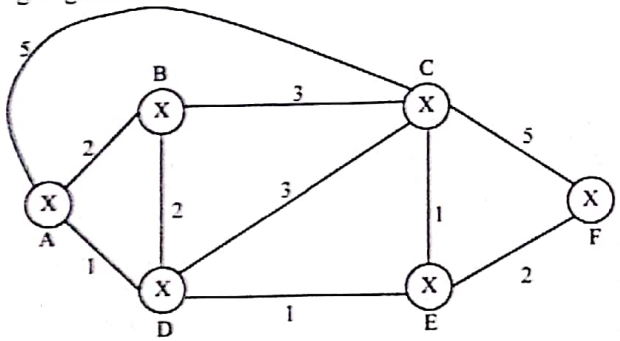
SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Briefly discuss about client-server architecture. Discuss about the major challenges of P2P architecture. (12)
   b) "Socket is analogous to door" – Justify the statement. What transport service does an app need? (08)
   c) What are the components of cookies? Describe how cookies keep state using database with proper examples. (10)
   d) FTP control connection is "out of band". Explain. (05)
2. a) Let, host at *abc.poly.edu* wants IP address for *defg.cs.uxyz.edu*. Show, how it is done using recursive queries for various DNS servers. (09)
   b) Explain the necessity of mail access agent in email system. Compare and contrast the functionalities of POP and IMAP. (10)
   c) Consider the following 20-byte header (IPv4). Find the header checksum. Why does IPv4 header checksum cover only data portion? (10)

4	5	0	28
1		0	0
4	17	0	
10.12.14.5			
12.6.7.9			

   d) How does conditional GET command improve performance of web proxy and local cache mechanism? (06)
3. a) Explain how DHCP works. What are the main benefits of DHCP? (09)
   b) Apply fragmentation on following information for IPv4 datagram. (13)
      i) Original Datagram: 5000 bytes, Header: 20 bytes.
      ii) MTU: 1st level 2000 bytes & 2nd level: 800 bytes.
   c) An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to 3 groups of customers: (13)
      i) 1st group: 52 customers, each needs 128 addresses.
      ii) 2nd group: 40 customers, each needs 256 addresses.
      iii) 3rd group: 100 customers, each needs 64 addresses. How many addresses are still unallocated? Note down the network addresses of each group.
4. a) Briefly explain how PAT works. Why is PAT used over NAT? (10)
   b) Determine the routing table/forwarding table at the router B from the following topology using a Link-state Routing Algorithm. (17)



   c) Concisely describe the error reporting messages of ICMP. (08)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Describe the working principle of Ethernet's CSMA/CD protocol with a detail look on exponential back-off algorithm. (10)
   b) Write filtering/forwarding algorithm of a switch. List the actions with corresponding conditions that a switch may perform when it receives a frame. (09)
   c) What is multiprotocol label switching (MPLS)? How does it work? Describe MPLS functionality with a simple example. (12)
   d) Describe silly windows syndrome created by the sender. (04)

6. a) How does TCP handle congestion? Describe with proper state diagram and graph. (15)
   b) Write down the connection establishment and half close in TCP. (10)
   c) How retransmission timeout is calculated in TCP? In which situation Karn's algorithm is applied? (10)

7. a) Briefly explain how Secure Socket Layer (SSL) works on top of TCP and what services it provides. (10)
   b) What is the difference between RDT 2.2 and RDT 3.0? Describe the FSM of RDT 3.0. (12)
   c) Write short notes on (i) Persistent Timer and (ii) Keepalive Timer. (08)
   d) What should be the relation between window sizes and sequence number in case of Selective Repeat? Why do we need the relationship? (05)

8. a) What are the shortcomings of symmetric key cryptography? How are those solved in public key cryptography. (06)
   b) In symmetric key system, how do two entities establish shared secret key over a network? Explain a solution with suitable example. (10)
   c) How does Multiple Access with Collision Avoidance (MACA) solve hidden terminal problem in wireless network? Explain with example. (07)
   d) Suppose, A wants to send a packet to B. If A does not know B's MAC address then how does it determine this MAC address in the following two situations: (12)
      i) A and B are in the same network.
      ii) A and B are in the different network.